FEBAUTO Racing

feb_racer: the memory racer

What	a lap to map the track, then a raceline driven by pure pursuit
Where from	the RoboRacer League 6 entry's mapping and raceline, on the car's contract
Status	scored practice 2026-10-24: loop 138.8 s and porto 110.8 s, clean, against the gap follower's 17

Formula Electric at Berkeley
September 2026, platform release v0.1.0

https://github.com/FEBAutonomous/feb-racing
https://febautonomous.github.io/feb-racing/

Contents

1 What it does 2

2 What it is not 2

3 Running it 2

4 The numbers 2

5 What the scored runs taught 3

6 Known limits 3

1 What it does

Three phases, visible in `./feb logs`:

Phase	What happens
WARMUP	the gap follower drives one lap at `warmup_speed` while a scan matcher builds an occupancy map in the start frame, correcting the odometry as the map fills. The lap closes when the car is back within `close_radius_m` of where it started, heading the same way, after `min_lap_m`.
COMPUTE	the map is frozen. A thread extracts the corridor centre, optimises a minimum-curvature line inside the corridor and gives it a friction-ellipse speed profile (`v_max`, `ay_max`, `a_acc`, `a_dec`). Under a second; the car keeps driving reactively meanwhile.
RACE	every scan is matched against the map. A pure-pursuit tracker aims at the raceline point one lookahead ahead and commands the profile's speed over the next `speed_horizon_m`. The gap follower's clearance brake stays armed underneath.

When the map match drops below `lost_score` for `lost_handoff_s`, the car hands back to the gap follower at `recover_speed` and takes the line back when the map matches again. The reactive law has no pose to lose, which is why it is the safety net; that idea comes straight from the league entry. After two seconds lost, or whenever the dead-man re-engages, it searches the whole map for itself once.

2 What it is not

It is not the league entry copied over. That node was 850 lines tied to the league's throttle commands and encoder ticks, with a sampling MPC and a dozen detectors for the league simulator's quirks. Kept: the scan matcher (`localizer.py`) and the raceline generator (`raceline_gen.py`), unchanged. Replaced: the controller, by pure pursuit on the contract's speed and steering angle; the inputs, by `/scan` and `/odom`; the detectors, by the dead-man. The racer is `racer.py`, under 300 lines.

3 Running it

```
./feb up --driver feb_racer                          # simulator, it drives from the
    start
./feb up --deadman --driver feb_racer                # hold SPACE: warm-up, then it races
    ; let go, it stops
./feb practice --driver feb_racer                    # scored: the warm-up lap is the
    rules' warm-up lap
./feb up --car --driver feb_racer --mode-name inspection   # the car: RB held, slow
    numbers from driver.car.yaml
```

The map lives in the node for as long as it runs. Stop and start the stack and it maps again.

4 The numbers

`config/driver.yaml`, every value commented. The ones to touch first:

Value	Meaning
`warmup_speed`	the mapping lap; slow is fine, the map only needs to be complete
`v_max`, `ay_max`	the speed profile's cap and lateral limit; the first thing to raise on a track it laps cleanly
`speed_scale`	scales the whole profile; 0.8 for a first run on a new track
`wall_margin_m`	how far the line keeps from the walls on top of half a car
`lookahead_gain`	pure pursuit's lookahead per m/s; higher is smoother and cuts corners more
`lost_score`	below this the pose is not trusted

`config/driver.car.yaml` is loaded on top on the car: slower warm-up, `v_max` 2.5, scale 0.8, and a longer `range_max` because the S2 sees further than the simulator's lidar.

5 What the scored runs taught

The free runs were clean from the first evening; the scored practice, at the league's 10 Hz, was not, twice. Both causes were measured, not guessed. First, our odometry's heading integrates the gyro and at 10 Hz drifts 18 degrees in 30 s and 84 by the end of a run; the racer now takes its heading from the IMU's absolute orientation when there is one. Second, with a good pose the car still clipped walls: at 10 Hz and a quarter-second actuator delay, pure pursuit at 4 m/s steers late. The tracker now aims from where the car will be one delay later, keeps 0.25 m from the walls and caps at 3 m/s. The same heading drift is why the cone racer lost localisation at 10 Hz; the house adapter will get the IMU heading too.

6 Known limits

- Walled tracks only. Cones are a different problem; see `feb_cone_racer`.
- A respawn or a carried car breaks the map frame until the global search finds it again.
- The car has not run it. The matcher and the map were built in the simulator with its lidar; the S2 is denser and sees further, which only helps, but nobody has watched it.
- Pure pursuit is a looser tracker than the league's MPC; the lap times have room.